CYBERTHORN 26’

MINDPULSE

A Privacy-Safe Predictive Cyber-Physical System for Classroom Wellbeing and Cognitive Load Monitoring

Problem Statement:

Learning environments often degrade before educators can visibly recognize the problem. Factors such as rising air quality deterioration, prolonged mental fatigue, excessive noise variability, and environmental discomfort can significantly reduce cognitive performance and increase stress — often *before* outward signs appear.

Existing solutions rely on cameras, wearables, or post-assessment feedback, which raise privacy concerns and react too late. There is a need for a **real-time, ethical, and non-intrusive system** that can identify early warning signs of suboptimal learning conditions and enable timely intervention.

Proposed Solution:

MindPulse is an Arduino-based **Cyber-Physical System (CPS)** that continuously monitors **anonymous environmental signals** to estimate **classroom-level overload risk** and **predict** *when* learning conditions are likely to degrade, enabling early & timely intervention. (EG: “Cognitive overload expected in ~12 min if conditions continue.")

MindPulse computes two indicators: a **Stress Index** (SI) for agitation and discomfort, and a **Cognitive Load Index** (CLI) for mental fatigue. Using **adaptive EMA-based calibration** and a **3-mode classroom system** (Lecture, Group, Exam), the system adjusts to different rooms and teaching contexts while minimizing false alarms.

Beyond alerts, MindPulse includes limited, low-risk **automation** such as activating ventilation during poor air quality and adjusting lighting when needed, while critical scenarios trigger a buzzer that alerts the teacher to prompt immediate human intervention. The system is **privacy-preserving** by design, with no personal, audio, or visual data collection.

System Architecture / Workflow:

1. **Sensing Layer (Physical System):** MindPulse uses low-cost, non-invasive sensors to collect environmental data only:

- Sound fluctuation (noise variability, not audio)
- Motion density (restlessness via PIR sensor)
- Air quality trend (CO_2 proxy)
- Temperature and humidity
- Ambient light level

No personal, audio, or visual data is collected.

2. **Cyber Layer (Processing, Calibration & Prediction)**
    - **Adaptive Calibration:** Performs an initial calibration to learn classroom baselines and continuously adapts using **EMA**, responding to deviations rather than fixed thresholds to reduce false alarms
    - **Context Awareness:** Operates in three selectable modes (**Lecture, Group, Exam**) with mode-specific sensitivity, distinguishing productive activity from overload.
    - **Prediction & Alerts:** Tracks trends in **Stress Index (SI)** and **Cognitive Load Index (CLI)** to estimate **time-to-risk** and trigger alerts only when conditions persist across multiple signals.
3. **Actuation & Feedback Layer**
    - RGB LEDs indicate real-time classroom state (Normal / Warning / Critical)
    - LCD displays show indices, predicted time to risk, and suggested actions
    - **Automatic ventilation fan activation** when CO_2 proxy exceeds adaptive thresholds
    - **Automatic lighting activation** when ambient light is insufficient
    - **Buzzer alert** triggered only during **critical predicted overload conditions**
    - Enables teachers to intervene early (short breaks, ventilation, pacing adjustments)

Innovation / Uniqueness:

- Predicts **future overload**, not just current conditions
- Uses **environment-based inference** instead of personal monitoring
- Fully **privacy-safe** (no cameras, no wearables, no data storage)
- Combines education, wellness, and CPS principles in a single system
- Low-cost and deployable in real classrooms
- Combines **predictive analytics with selective automation**, moving beyond passive monitoring
- Uses **adaptive EMA calibration** instead of fixed thresholds

Hardware and Software Technology Stack:

Hardware:

- Arduino Uno
- DHT22 (temperature & humidity)
- MQ-135 (air quality / CO_2 proxy)
- Sound sensor (e.g., KY-038)
- PIR motion sensor (eg: HC-SR501)
- LDR (light sensor)
- Mode selection push-button
- RGB LEDs
- LED NeoPixel Strip
- Piezo buzzer
- Ventilation/ Exhaust Fan
- LCD displays

Software:

- Arduino IDE
- Embedded signal processing & predictive algorithms
- EMA-based adaptive calibration logic & Mode-aware thresholding (Lecture / Group / Exam)
- No cloud dependency or external servers



[IMAGE GENERATED USING AI ONLY FOR VISUAL DEMONSTRATION PURPOSES]

Target Users / Beneficiaries:

- School and university students
- Teachers and invigilators
- Educational institutions seeking smart, ethical classroom solutions

Expected Impact:

- Early detection of suboptimal learning conditions
- Improved classroom engagement and learning outcomes
- Reduced burnout and anxiety during long or intensive sessions
- Supports UAE's vision for smart, **student-centric education environments**

Feasibility & Scalability:

- Easily buildable using off-the-shelf components
- Scalable to libraries, exam halls, and study spaces
- Can be integrated with future dashboards or building management systems

Ethics & Security:

- No personal data collection
- No audio/video recording
- Collective, anonymous sensing only
- All processing occurs locally
- **Human-in-the-loop control:** automation is limited to low-risk environmental awareness, while all educational decisions remain with the teacher
- Ethical by design and compliant with privacy-first principles